DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Course: CSE489
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Project Title: Service Booking App

Project Features:

1. User Account & Authentication
   - Role-Based Access Control (RBAC): Three distinct user roles (Customer, Service Provider, Admin) with restricted routing.
   - Secure Authentication: JWT (JSON Web Token) session management and bcrypt password hashing.
   - Profile Management: View and edit full name, phone number, and password.
   - Role-Specific Navigation: Custom bottom navigation bars tailored to the user's role (e.g., Customers see "Services/Profile", Providers see "Dashboard/Profile").

2. Service Discovery (Customer Features)
   - Service Browsing: View a scrolling feed of all available services.
   - Category Search/Filtering: Filter the available services by category using a search dialog.
   - Detailed Service Pages: View high-resolution service images, descriptions, pricing, duration, and provider contact info.

3. Booking & Timeline System (Customer Features)
   - Date Selection: Native date picker for choosing appointment days.
   - Visual Booking Timeline: A dynamic, 3-step progress bar (Booked → Pending → Completed) tracking the status of each appointment.
   - Booking Cancellation: Ability to cancel a pending booking, which automatically alerts the provider.
   - Service Confirmation: "Got Service" button to mark a pending booking as completed once the service is rendered.
   - Status Badges: Color-coded UI badges representing booking states (Pending, Confirmed, Completed, Cancelled) and payment states (Paid, Refunded).

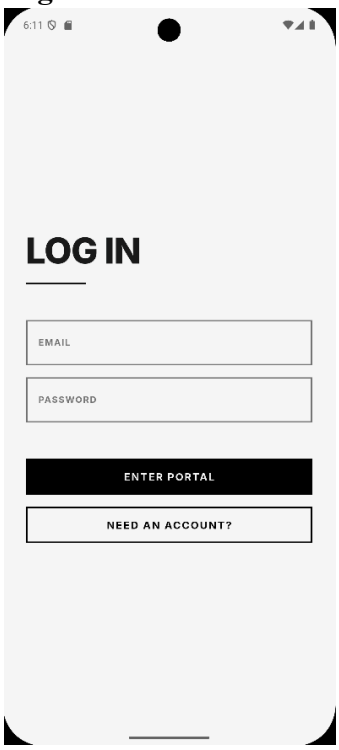
4. Payment Integration
   - Stripe Secure Gateway: Native Stripe Payment Sheet integration for collecting card details without storing sensitive PCI data.
   - Instant Payment Confirmation: Bookings are only finalized upon successful Stripe charge.

- Automated Refunds: Providers can issue refunds directly from their dashboard, which communicates with Stripe's API to reverse the charge.
- Invoicing: Digital transaction receipts available for customers showing service details, amount paid, and Stripe confirmation.

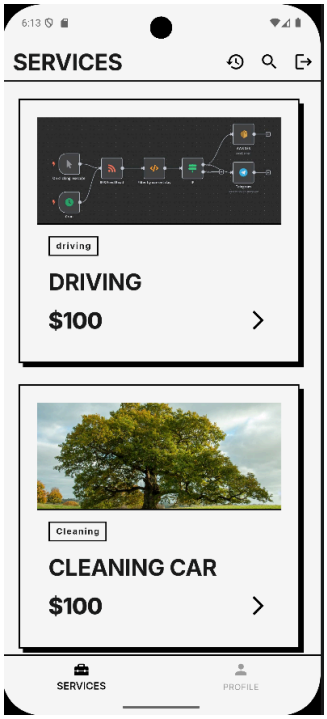
5. Service Provider Dashboard
    - Service Publishing (Multipart Uploads): Providers can create new service listings and upload images directly from their phone gallery using multer.
    - "My Services" Management: A dedicated tab showing all active services published by that specific provider.
    - Incoming Booking Management: A dedicated tab to view all bookings made by customers, including customer contact info and payment status.
6. Notifications & Reminders
    - Provider Inbox: In-app notification system (bell icon) that alerts providers instantly when a customer creates or cancels a booking.
    - Customer Push Reminders: Device-level push notifications (using flutter_local_notifications) scheduled automatically to alert the customer exactly 2 hours before their booked appointment.
7. Admin Control Panel
    - User Management: View all registered customers and providers, with the ability to instantly Ban/Unban accounts.
    - Global Booking Oversight: Monitor every transaction on the platform, including customer info, service booked, and payment status.
    - Service Moderation: View all published services and permanently delete inappropriate listings.
    - Revenue & Analytics Dashboard: Real-time calculation of platform statistics, including total active users, booking status breakdown, total revenue, total refunded, and net platform revenue.
8. UI/UX Design
    - Brutalist Design System: A cohesive, custom design language featuring sharp corners, thick borders, high-contrast colors, and bold typography.
    - Custom Widgets: Reusable UI components (BrutalistCard, BrutalistButton, BrutalistTextField) ensuring visual consistency across the entire application.

Screenshots:

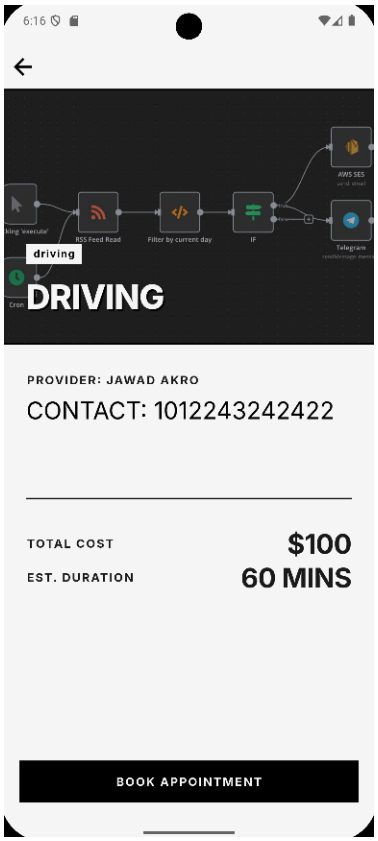
Login



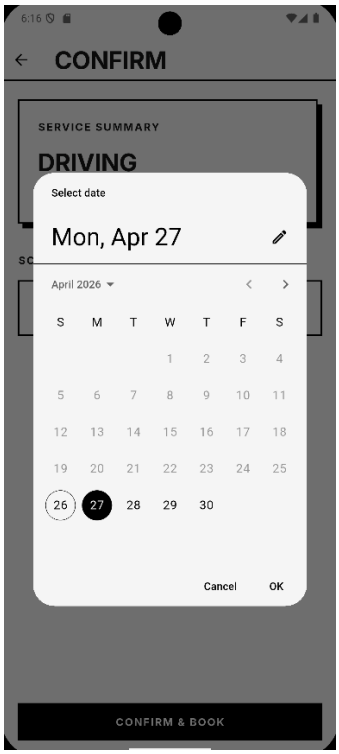
Service Available for Customers:



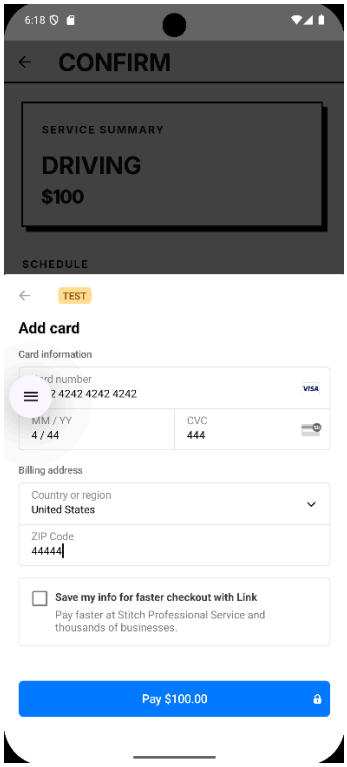
Service Tab for the customer:



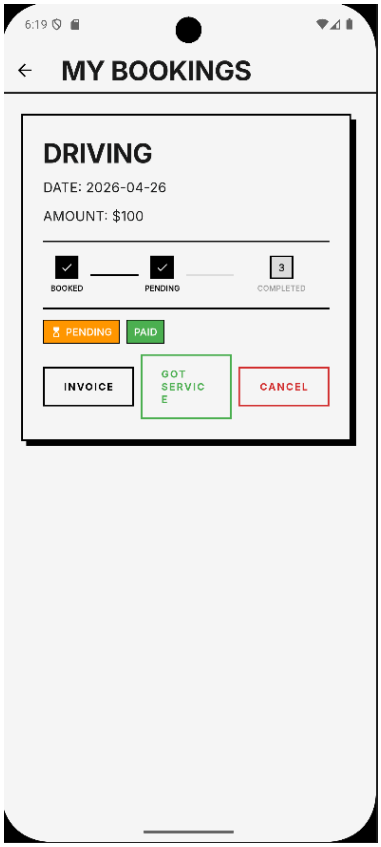
Date Selection for Service:



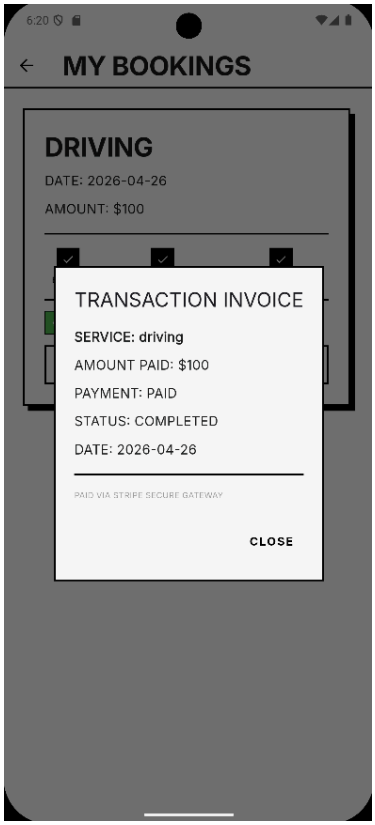
Stripe Payment Window:



My Booking for Customers:

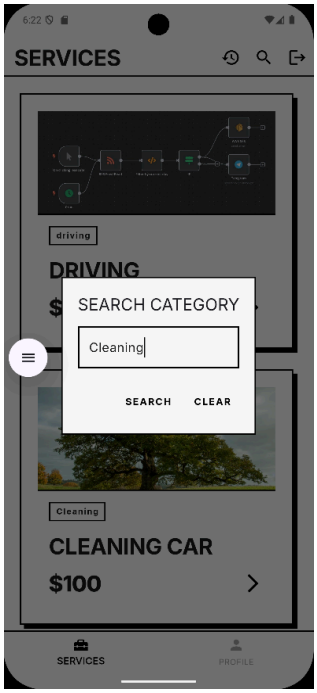


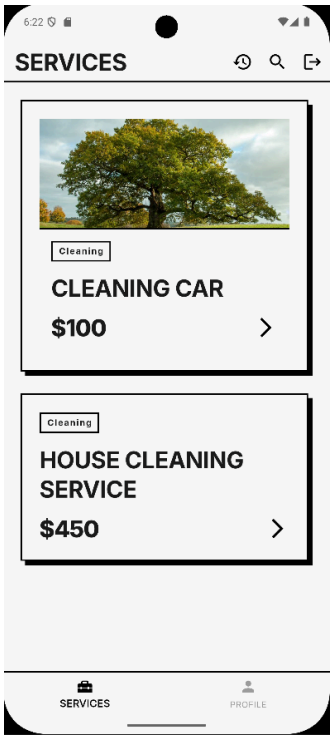
Invoice After Payment :



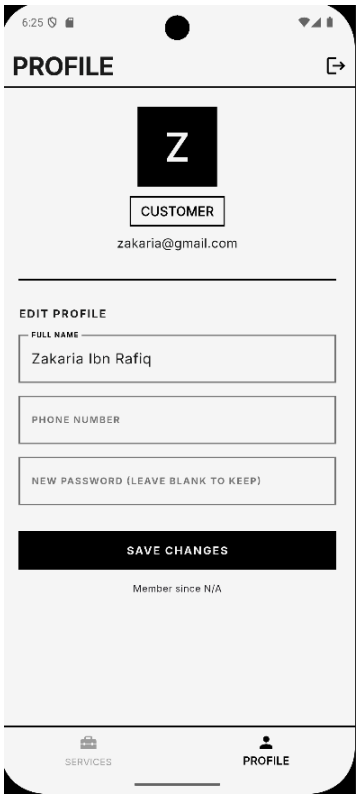
After confirmation of the customer's service invoice is generated.

Search Functionality:

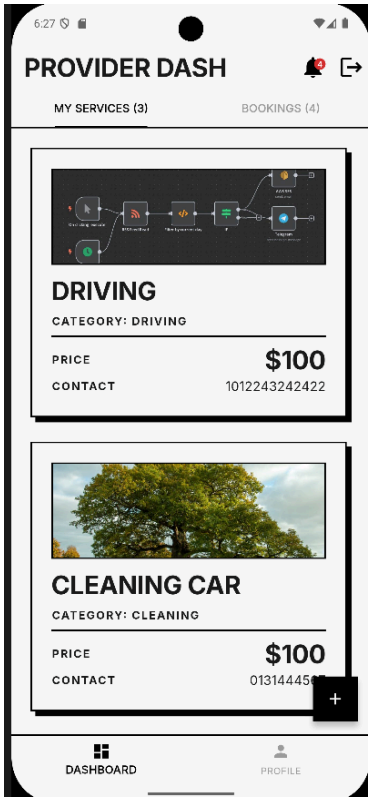




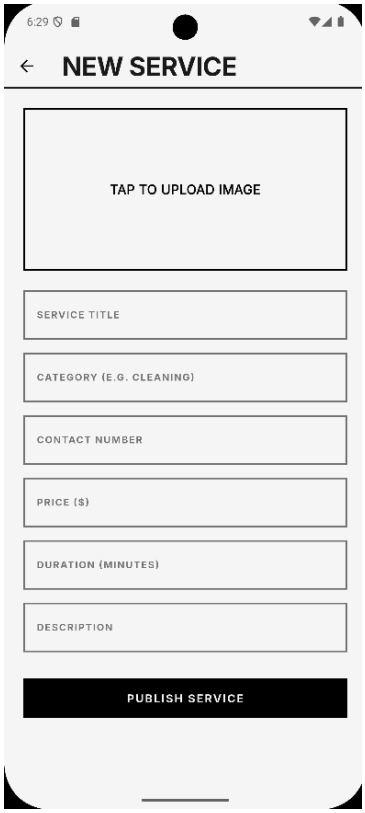
Customer Profile :



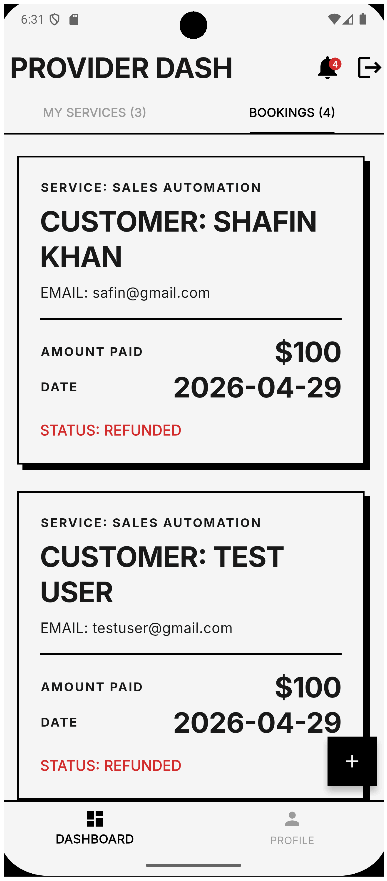
Service Provider's Dashboard:



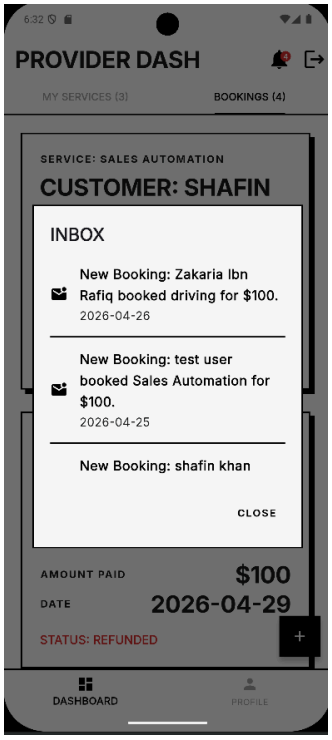
New Service Upload:



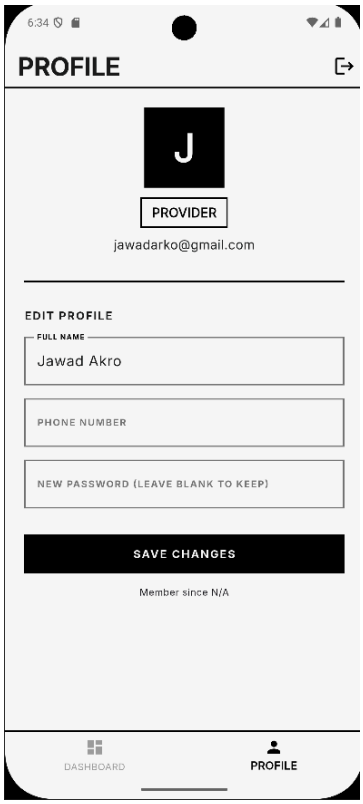
Booking Received By Service Providers:



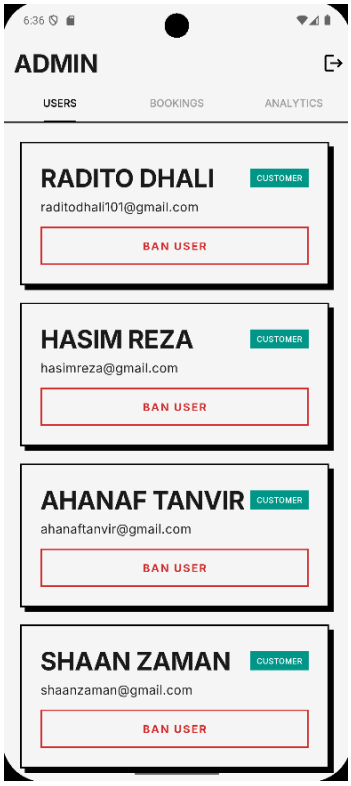
Notification System for Service Providers:



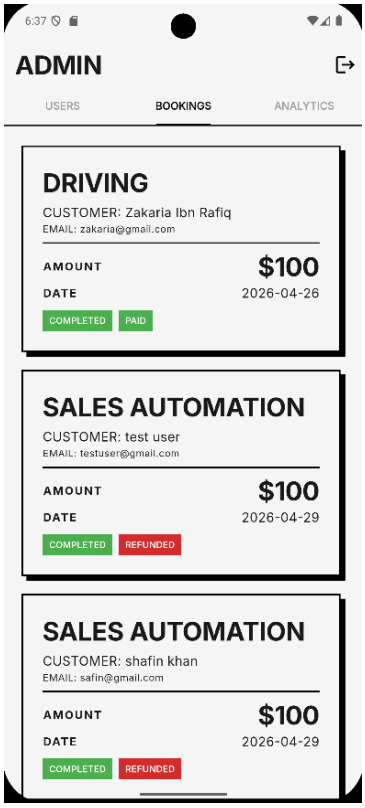
Profile For Service Provider:



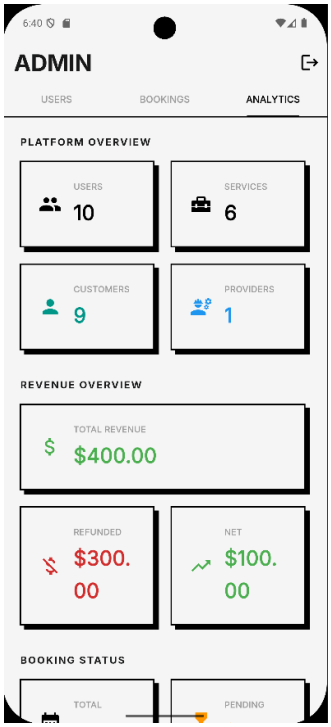
Admin Dash for All Users:



Services Book Dash for Admin:



Analytics Dash for Admin:



Online Resources used:

a) *Reference:*

- *Youtube videos:https://youtube.com/playlist?list=PLFyjjoCMAPtzn7tFLRV3eny7G74LnlMRt&si=k1HQkZ6E-G3UVhHn,*
  Flutter Stripe Payments Tutorial | Accept Payments Within Flutter Applicati…

Future Enhancements:

Following enhancement can be added to the current system which will improve the system.

- Real-Time Calendar Integration & Time Slots: Synchronize provider availability with a native calendar view.
- In-App Chat System: Real-time communication gateway between customers and providers using WebSockets.
- Post-Service Ratings and Reviews System: Trust-building mechanism featuring star ratings and text feedback.
- Social Logins: OAuth2 integration for rapid account creation via Google, Apple, and Facebook.
- OTP Phone Number Verification: Enhanced security through mobile number validation.
- Password Recovery / Reset Flow: Secure automated credentials recovery via email tokens.
- Email & SMS Notifications: Automated alerts powered by Twilio and SendGrid APIs.
- Location-Based Search: Integration of Google Maps SDK for localized service discovery.
- Live Location Tracking: Real-time GPS monitoring for active provider service transit.
- Automated Platform Commission Fee Splitting: Programmatic revenue distribution between providers and the platform.
- Multi-Language & Multi-Currency Support: Localization and internationalization for global scalability.